

Laboratory Activity 5

Addition and Subtraction

OBJECTIVES:

At the end of the exercise, the students should be able to:

- create an assembly program that performs addition and subtraction;

SOFTWARE REQUIREMENT:

- EMU8086 4.08

PROCEDURE:

1. Create a new COM template.
2. Use the MOV instruction to assign 4 to AL. Then, assign 8 to BL.
3. Add the value of BL to AL by typing **add bl, al** then proceed to the next line.
4. Sub the value of BL by typing **sub bl, 1** then next line.
5. Type **mov cx, 8** to be able to display the sum in its binary equivalent.

6. Group the following instructions and name it **display**.

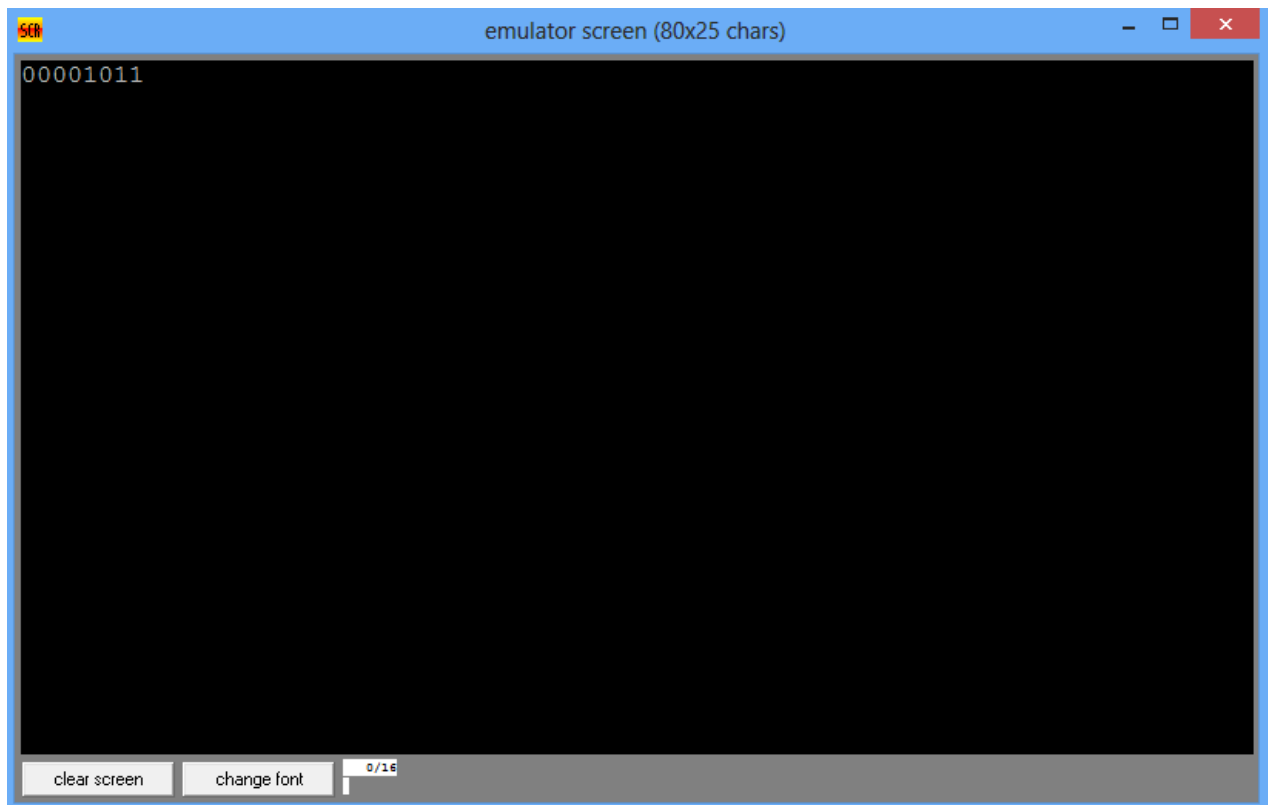
Instruction	Description
mov ah, 2	Print function
mov dl, '0'	Prints 0
test bl, 10000000b	Tests first bit
jz zero	Jumps if zero
mov dl, '1'	Prints 1

7. Group the following instructions and name it **zero**.

Instruction	Description
int 21h	Ends the current process
shl bl, 1	Shifts all bits left

8. Type **loop display** then press Enter. This will print each digit until it reaches eight (8).
9. Add the two (2) instructions to terminate the program upon pressing Enter: **mov ah, 0** and **int 16h**.
10. Emulate the finished program.

Sample Output:



LABORATORY WORK

1. Create a new COM Template. Write an assembly program that will display the difference of two (2) decimal numbers in binary.
2. Save the program as **LabAct5.asm**.
3. Upload your video recording and screenshots to the submission bin.

